Metamorphic Rocks

- Altered by heat and pressure
- Not completely melted
- Special minerals form
- As with all rock types, there are a dizzying variety of classes and names (Greenschist, blueschist, eclogite, etc)

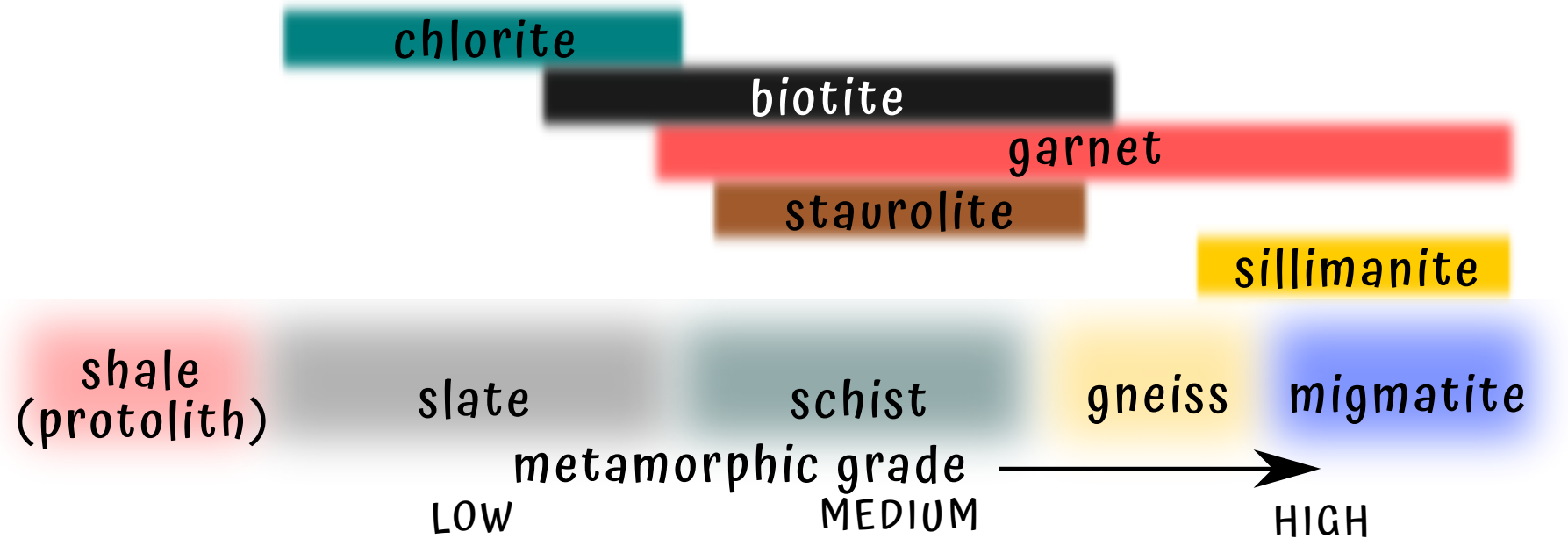
Types of Metamorphism

- Contact metamorphism (e.g. metamorphic halo around igneous intrusion)
- Regional metamorphism (large tectonic events)

Metamorphic Grade

Metamorphic minerals and the different rock types form at different temp and pressures.

- Chlorite (low T and P)
- Garnet (intermediate T and P)
- Staurolite (high T and P)
- Sillimanite (high T and P)



Metamorphic Rock Classification

Foliated Rocks

- Slate (fine grained, dull, planar breaks)
- Phyllite (fine grained, shiny, planar breaks)
- Schist (course, clear orientation of minerals)
- Gneiss (minerals segregate into bands)

No-Foliated Rocks

- Quartzite
- Marble
- Hornfels

Other Textures

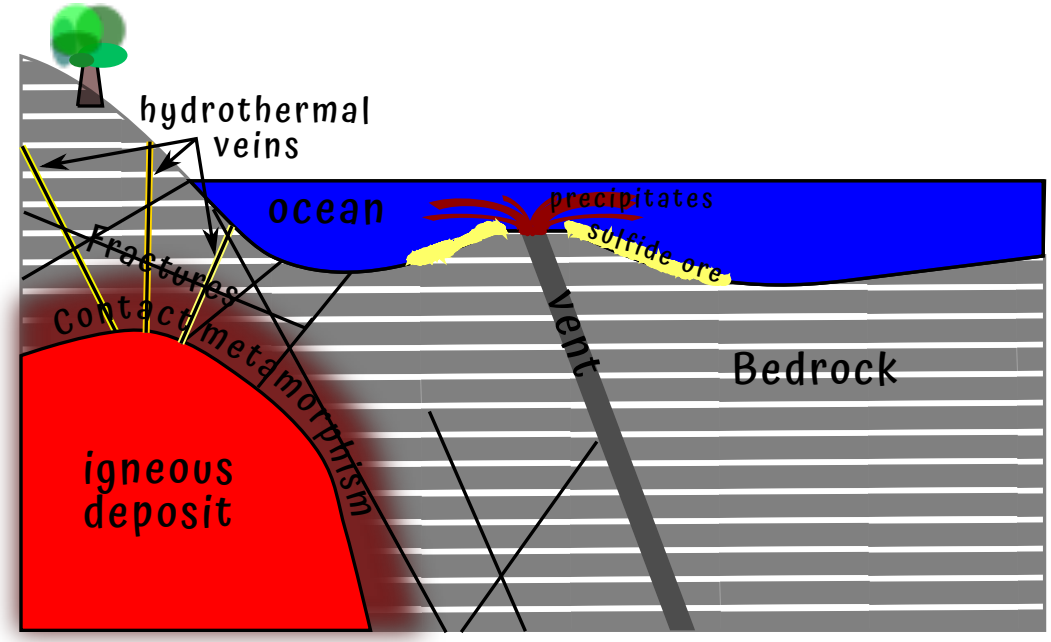
- Migmatite

garnet-sillimanite gneiss

Migmatite Example

Mineral resources

- Hydrothermal deposit: fluid migration from magma carrying leached metals and re-deposits metals in surrounding rocks (mineral veins)
  - High temp and pressure mobilizes metals.
  - Can change the mineralogy
  - Deposits enriched in copper, silver and gold (and others)
- Other 'valuable' economic minerals and rocks
  - Graphite
  - Talc
  - Asbestos (health risk)
  - Garnet (gemstone, abrasive)
  - Marble
  - Slate



Metamorphic Rocks: Applications

- Foliation creates planes of weakness, permeability
  - Blasting/tunneling
  - Slope stability
- Alkali-silica reaction (fine grained qtz-rich)

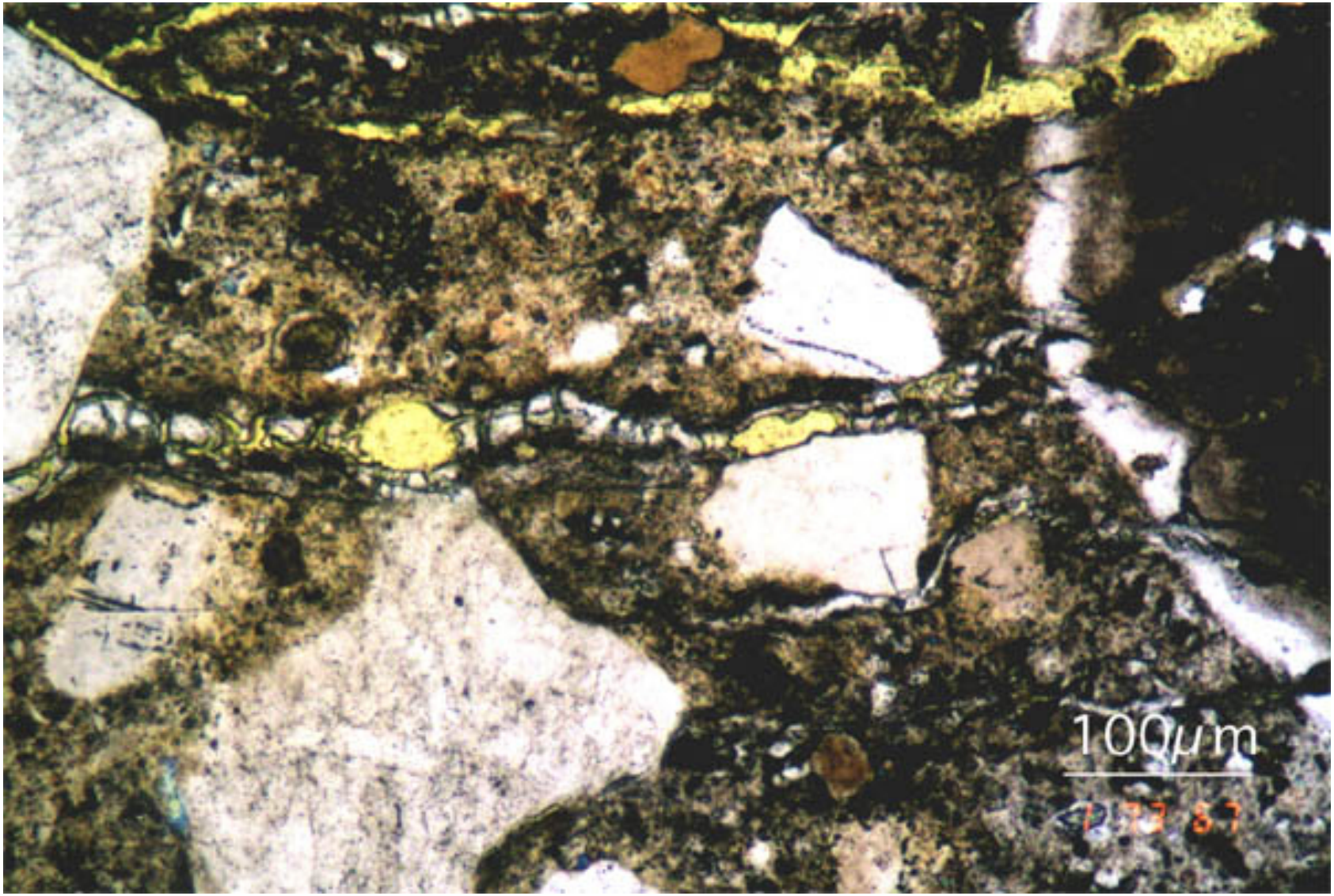
$$Ca(OH)_2 + H_4SiO_4 \rightleftharpoons CaH_2SiO_4 \cdot 2H_2O$$

Expansive gel forms after cement sets

- Weakness in micas/water penetration (freeze-thaw)
- Mica reaction with water (in schist & gneiss) can impact concrete
- Crushed foliated rocks can yield angular fragments, used for aggregate (phyllite, gneiss); however flat fragments cause mixing issues, directional properties.

Alkalii Mineral Reaction

Chert grain (black object, right side) in polarizing micrograph. Alkalii silica gel in cracks.



Study Guide

1. How are metamorphic rocks classified?
2. What are the different names for foliation? What is foliation? Why is it important?
3. What is a metamorphic grade? How are is a grade determined?
4. How to metamorphic rocks impact construction activities (excavation and concrete)?
5. How does metamorphism create economically valuable deposits?